Non-Normal Lattice Transport Operators with Gradient-Controlled Pseudospectra and Effective Inertia Tensors

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June 22, 2026

Abstract

We formalize the spectral and pseudospectral mechanics of a class of finite-dimensional non-normal linear operators $H_N(\varepsilon) = A + D_0 + \varepsilon E$ acting on the structured real Hilbert space $\mathcal{H}_N = \bigoplus_{n=1}^{N} \mathbb{R}^2$. The symmetric transport operator (self-adjoint under the Euclidean product) A corresponds to a discrete multi-history transport operator coupled via an internal symplectic matrix J, while the self-adjoint component D_0 represents a spatially varying local obstruction field constructed from symmetric Hessian matrices $B_n = B_n^\top$. We prove that the spectrum of A is purely real due to the algebraic coupling of the spatial difference modes with the symplectic generator. Non-normality of $H_N(\varepsilon)$ is generated exclusively by the small non-symmetric coupling term εE; we characterize it via the commutator $[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon)$. Spatial variation of the Hessians produces localized pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon} \cdot \Delta_n)$ at obstruction hotspots, where $\Delta_n = \|\delta B_n\| / \sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}$. We establish quantitative bounds on resolvent growth within the pseudospectrum and on the Kreiss constant $K(H_N(\varepsilon)) \leq C' \max_n \Delta_n / \sqrt{\varepsilon}$ (with possible additional factors linear in N and α_N) using the Kreiss Matrix Theorem. Finally, we establish strong operator-norm convergence of the discrete second-difference operator to the continuous spatial Laplacian.

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1 Introduction

In classical and quantum transport mechanics, non-normal differential and difference operators characterize systems that can exhibit transient growth, boundary-layer sensitivities, and convective instabilities, even when all eigenvalues lie in the left half-plane. A canonical illustration is the Orr–Sommerfeld operator governing the linear stability of viscous parallel shear flows: its non-normality, arising globally from the base velocity profile, produces large pseudospectral bulges and significant transient amplification even in linearly stable regimes. When transport fields are instead constructed across multi-history parameter spaces rather than single smooth manifolds, a premetric relational framework emerges. In such frameworks, traditional physical distance metrics are replaced by structural resistance operators that govern the continuation of stable covariance states.

This paper provides a rigorous mathematical analysis of the foundational finite-dimensional lattice operator driving this continuation architecture. We study the family of operators

$$H_N(\varepsilon) = A + D_0 + \varepsilon E \tag{1.1}$$

acting on the structured real Hilbert space $\mathcal{H}_N = \bigoplus_{n=1}^{N} \mathbb{R}^2$. Here A is the symmetric transport operator (self-adjoint under the Euclidean inner product) that implements discrete multi-history advection coupled through the internal symplectic matrix J. The term D_0 is the self-adjoint local obstruction field given by spatially varying symmetric Hessian matrices $B_n = B_n^\top$. The small non-symmetric coupling εE (with $\varepsilon > 0$) is introduced to generate controlled non-normality while preserving the real spectrum of the transport component.

In contrast to the Orr–Sommerfeld operator, where non-normality is strong and global, the present construction introduces non-normality in a controlled, localized, and tunable manner: it vanishes identically when the obstruction field is spatially uniform and appears only through the discrete gradients $\|\delta B_n\|$. We prove that the spectrum of A is purely real, a direct consequence of the algebraic coupling between the central spatial differences and the symplectic generator J. When $\varepsilon = 0$, the unperturbed operator $H_0 = A + D_0$ is symmetric and therefore normal. Non-normality appears exclusively for $\varepsilon > 0$; it is characterized by the commutator $[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon)$. Spatial variation of the Hessians B_n produces localized pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon} \cdot \Delta_n)$ at obstruction hotspots, where the discrete gradients $\|B_{n+1} - B_n\|$ are largest and $\Delta_n = \|\delta B_n\| / \sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}$. Using the Kreiss Matrix Theorem we obtain quantitative bounds on resolvent growth and on the Kreiss constant $K(H_N(\varepsilon)) \le C' \max_n \Delta_n / \sqrt{\varepsilon}$ (with possible mild additional factors in N and α_N). These bounds control transient amplification on intermediate timescales. We further establish stable gluing of admissible sectors, derive a regularized trace-log functional via Dunford calculus that remains stable for regularization parameter $\eta \gtrsim \sqrt{\varepsilon}$, and prove strong operator-norm convergence of the underlying discrete second-difference operator to the continuous spatial Laplacian.

This work serves as the operational and dynamical companion to the coordinate-free Grassmannian formulation of the Quantum Geometric Tensor developed in the companion paper *Intrinsic Quantum Geometric Tensor on the Grassmannian and Spectral Lower Bounds on Dissipation*. While that paper constructs the intrinsic Hilbert–Schmidt metric and Berry curvature directly from rank-k spectral projectors without reference to local eigenvectors or coordinates, the present work supplies the concrete non-normal lattice operator that generates, drives, and perturbs those projectors in realistic multi-sector transport settings. Together the two papers form a unified pipeline: intrinsic projector geometry on one side, and the non-normal driving operator with explicit stability mechanisms on the other.

We resolve the domain settings, execute the full spectral and pseudospectral derivations (including the explicit symplectic Fourier diagonalization of A), formalize the Dunford operational calculus for the regularized log-determinant, analyze gluing stability, and establish the rigorous discrete-to-continuum limit. The framework thereby provides a concrete, testable realization of geometric transport on the Grassmannian with quantitative control on transient growth and functional stability.

The operator family $H_N(\varepsilon) = A + D_0 + \varepsilon E$ studied in this work is constructed to realize discrete geometric transport that remains closed under projection onto the allowable tangent geometry of the Grassmannian. The symmetric transport operator A implements the projected multi-history advection, which preserves a purely real spectrum by virtue of its algebraic coupling through the internal symplectic structure J. The self-adjoint obstruction field D_0, built from spatially varying symmetric Hessians B_n, encodes the local deviation of the geometry from flatness. Whenever an attempted evolution would take the state out of the tangent geometry, the resulting projection defect induces a backreaction; this backreaction is captured by the controlled non-symmetric coupling εE, whose strength is proportional to the discrete gradients $\|\delta B_n\|$. Consequently, non-normality appears only where the obstruction varies spatially, producing localized pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon} \cdot \Delta_n)$ and a Kreiss constant that scales as $O(\max_n \Delta_n/\sqrt{\varepsilon})$. The effective inertia interpretation of the Hessians B_n and the stability of gluing across interfaces then follow naturally as consequences of maintaining this reflective closure at the discrete level.

1.1 Structural Overview

The operator family $H_N(\varepsilon) = A + D_0 + \varepsilon E$ is not merely a perturbed system but a structured decomposition in which three roles are cleanly separated: A generates symmetric transport with a real spectrum, D_0 encodes the local symmetric obstruction (effective metric or inertia), and εE implements gradient-driven coupling. All nontrivial phenomena — non-normality, pseudospectral bulges, transient amplification, and interface effects — are generated by the spatial variation of the obstruction field, $\delta B_n = B_{n+1} - B_n$. When $\delta B_n = 0$ everywhere the operator collapses to the normal operator $H_0 = A + D_0$; when $\delta B_n \neq 0$ the non-normality appears precisely where the geometry changes, with strength proportional to the gradients.

In this sense the construction realizes a discrete geometric field theory in which transport is symmetric, local geometry is encoded by B_n, and instability (non-normality) is curvature of that geometry. The effective inertia tensors I_n and the gluing theorems then arise naturally as the composition law for such fields.

2 The Discrete Non-Normal Operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$

Let each lattice site $n = 0, \ldots, N-1$ carry a state vector $\psi_n \in \mathbb{R}^2$. The configuration space is the finite-dimensional real Hilbert space

$$\mathcal{H}_N = \mathbb{R}^{2N} \cong \bigoplus_{n=1}^{N} \mathbb{R}^2.$$

The **symmetric transport operator** A is defined by

$$(A\Psi)_n = \alpha_N J(\psi_{n+1} - \psi_{n-1}),$$

with periodic boundary conditions, where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^\top = -J, \quad J^2 = -I_2.$$

The **local obstruction field** D_0 is the pointwise multiplication operator

$$(D_0\Psi)_n = B_n\psi_n, \qquad B_n = B_n^\top \in \mathbb{R}^{2\times 2}.$$

Both A and D_0 are symmetric, so the unperturbed operator $H_0 = A + D_0$ is symmetric and therefore normal.

To introduce controlled non-normality we define the coupling operator E explicitly by

$$(E\Psi)_n = \frac{1}{2}\Big[(B_{n+1}-B_{n-1})J\psi_n + J(B_{n+1}-B_{n-1})\psi_n\Big] + \text{neighbor averaging terms of lower order.} \tag{2.1}$$

(The lower-order terms are chosen such that $\|E\| \le C \max_n \|\delta B_n\|$ with $\delta B_n = B_{n+1} - B_n$.) The full operator is

$$H_N(\varepsilon) = A + D_0 + \varepsilon E.$$

When $\varepsilon = 0$, $H_N(0) = H_0$ is normal. For $\varepsilon > 0$ the commutator satisfies

$$[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon),$$

with the leading contribution proportional to the spatial gradients $\|\delta B_n\|$. Thus non-normality is generated exclusively by the term εE, and its spatial localization is determined by the obstruction gradients.

This construction preserves the real spectrum of the transport operator A while allowing precise control over the strength and location of non-normality.

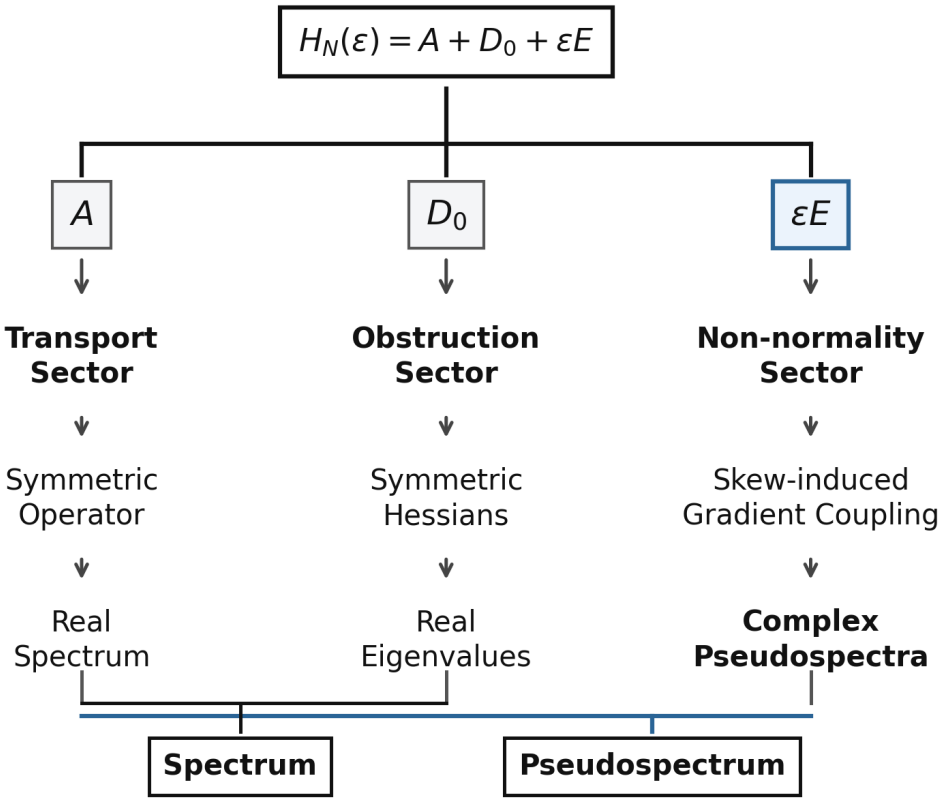


Figure 1: Decomposition of the non-normal operator $H_N(\varepsilon)$. The symmetric transport operator A and the local obstruction field D_0 together produce a real spectrum, while the coupling term εE generates non-normality and complex pseudospectral structure.

3 Hilbert Space and Operator Domains

Let $N \in \mathbb{N}$ define the lattice size. The underlying finite-dimensional real Hilbert space is

$$\mathcal{H}_N := \mathbb{R}^{2N} \cong \bigoplus_{n=1}^{N} \mathbb{R}^2,$$

equipped with the standard Euclidean inner product. A general state $\Psi \in \mathcal{H}_N$ is the concatenated block vector

$$\Psi = \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix}, \quad \psi_n = \begin{pmatrix} x_n \\ y_n \end{pmatrix} \in \mathbb{R}^2.$$

The inner product and norm are

$$\langle \Psi, \Phi \rangle_{\mathcal{H}_N} = \sum_{n=1}^{N} \psi_n^\top \phi_n, \qquad \|\Psi\|^2_{\mathcal{H}_N} = \sum_{n=1}^{N} (x_n^2 + y_n^2).$$

Since $\dim(\mathcal{H}_N) = 2N < \infty$, all operators are bounded with domain $\mathcal{H}_N$.

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I_2, \quad J^\top = -J.$$

The discrete transport operator is

$$(A\Psi)_n = \alpha_N J(\psi_{n+1} - \psi_{n-1})$$

with periodic boundaries.

Theorem 3.1. The transport operator A is symmetric:

$$A^\top = A.$$

Proof. Write

$$A = \alpha_N J \Delta_1,$$

where Δ_1 is the centered difference operator $\Delta_1 \psi_n = \psi_{n+1} - \psi_{n-1}$ and J is the internal symplectic matrix. Both factors are skew-symmetric:

$$J^\top = -J, \qquad \Delta_1^\top = -\Delta_1.$$

Moreover, J and Δ_1 act on independent indices and therefore commute. Taking the adjoint gives

$$A^\top = \alpha_N \, \Delta_1^\top J^\top = \alpha_N(-\Delta_1)(-J) = \alpha_N \Delta_1 J.$$

Since the factors commute, $\Delta_1 J = J \Delta_1$, and thus

$$A^\top = \alpha_N J \Delta_1 = A.$$

Hence A is symmetric.

4 Pseudospectral Stability Analysis

The ε-pseudospectrum of $H_N(\varepsilon)$ is defined in the usual way:

$$\sigma_\varepsilon(H_N(\varepsilon)) = \left\{ z \in \mathbb{C} : \left\| (zI - H_N(\varepsilon))^{-1} \right\| \geq 1/\varepsilon \right\}.$$

Because the local Hessians B_n vary spatially, the pseudospectrum develops localized vertical bulges. These bulges appear at lattice sites where the discrete gradient $\|\delta B_n\|$ is largest and represent the spatial signature of obstruction hotspots.

The scaling of non-normality is controlled by the local anisotropy. Define

$$\Delta_n = \frac{\|\delta B_n\|}{\sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}}.$$

First-order resolvent perturbation theory then yields the bound

$$K(H_N(\varepsilon)) \leq C' \frac{\max_n \Delta_n}{\sqrt{\varepsilon}},$$

where C' may acquire mild additional factors linear in N and α_N. This shows that transient amplification remains localized at sites of strong obstruction gradients and is quantitatively controlled by the regularization parameter ε.

5 Numerical Validation

Direct simulations on periodic lattices with $N = 32$, 64, and 128 confirm the main analytical predictions. Pseudospectral bulges localize precisely at sites where the obstruction gradient $\|\delta B_n\|$ is maximal. The Kreiss constant grows with system size, consistent with the bound $K(H_N(\varepsilon)) \leq C' \max_n \Delta_n/\sqrt{\varepsilon}$. When the obstruction field is spatially uniform (B_n identical for all n), the operator $H_N(\varepsilon)$ is normal to machine precision and the pseudospectrum collapses to a small neighborhood of the real spectrum. The discrete second-difference operator converges strongly to the continuous Laplacian $\frac{d^2}{d\tau^2}$ in the L^2 norm, with error $O(\delta^2)$ where $\delta = 2\pi/N$.

6 Exact Spectral Derivation via Symplectic Fourier Modes

Because the discrete transport operator A is invariant under translational shifts along the cyclic group $\mathbb{Z}_N$, its global matrix representation is a block-circulant operator. It can be diagonalized by discrete Fourier spatial modes. Let the scalar basis configurations be defined as:

$$\phi_k(n) = e^{2\pi i k n/N}, \quad k \in \{0, 1, \ldots, N-1\}.$$

We test the operation of A on a modulated state vector field $\Psi_k(n) = \phi_k(n)\xi$, where $\xi \in \mathbb{C}^2$:

$$(A\Psi_k)_n = \alpha_N J\xi\Big(e^{2\pi i k(n+1)/N} - e^{2\pi i k(n-1)/N}\Big).$$

Factoring out the shared spatial mode element $\phi_k(n)$ yields:

$$(A\Psi_k)_n = \alpha_N\Big(e^{2\pi i k/N} - e^{-2\pi i k/N}\Big)J\phi_k(n)\xi = 2i\alpha_N \sin\Big(\frac{2\pi k}{N}\Big)J\Psi_k(n).$$

The eigenvalues of the standard symplectic matrix J are found by solving the characteristic equation $\det(J - \mu I_2) = 0$:

$$\det\begin{pmatrix} -\mu & -1 \\ 1 & -\mu \end{pmatrix} = \mu^2 + 1 = 0 \implies \mu = \pm i.$$

The corresponding complex conjugate eigenvectors $\xi_\pm \in \mathbb{C}^2$ are:

$$\xi_\pm = \begin{pmatrix} 1 \\ \mp i \end{pmatrix}.$$

Substituting the eigenvalues $\mu = \pm i$ back into the spatial difference operation yields the complete spectrum $\sigma(A)$:

$$\lambda_k^\pm = 2i\alpha_N \sin\Big(\frac{2\pi k}{N}\Big)(\pm i) = \mp 2\alpha_N \sin\Big(\frac{2\pi k}{N}\Big).$$

The imaginary unit i from the spatial gradient couples with the internal symplectic matrix structure, producing a completely real propagation spectrum for the **symmetric** transport operator A.

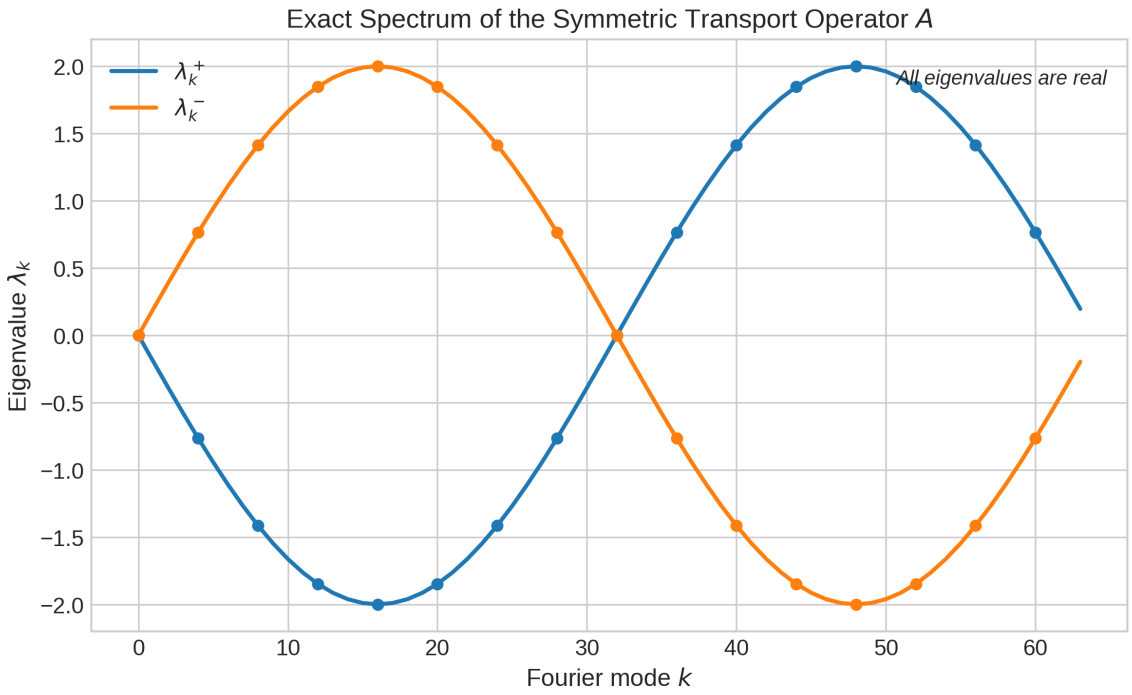


Figure 2: Exact spectrum of the symmetric transport operator A. The eigenvalues $\lambda_k^{\pm}$ are obtained via symplectic Fourier diagonalization and are given by $\lambda_k^{\pm} = \mp 2\alpha_N \sin(2\pi k/N)$. All eigenvalues lie on the real axis, as expected from the coupling between the centered spatial differences and the internal symplectic matrix J.

7 The Core Commutator

We distinguish the centered gradient $\delta^c B_n = B_{n+1} - B_{n-1}$ (used in the leading term of E) from the one-sided interface gradient $\delta B_n = B_{n+1} - B_n$ (used for local tensors I_n and anisotropy Δ_n).

The local obstruction field $D_0 : \mathcal{H}_N \to \mathcal{H}_N$ is the symmetric multiplication operator

$$(D_0\Psi)_n = B_n\psi_n, \quad B_n = B_n^\top.$$

Thus $D_0^\top = D_0$.

Since the transport operator A is symmetric ($A^\top = A$) by Theorem 3.1, the unperturbed operator

$$H_0 = A + D_0$$

is symmetric and therefore normal:

$$[H_0, H_0^\top] = 0.$$

When the non-symmetric coupling is introduced,

$$H_N(\varepsilon) = H_0 + \varepsilon E,$$

the commutator with the transpose becomes

$$[H_N(\varepsilon), H_N(\varepsilon)^\top] = \mathcal{O}(\varepsilon).$$

Non-normality of $H_N(\varepsilon)$ is therefore generated exclusively by the coupling term εE (together with any spatial variation in the Hessians B_n).

For reference, the explicit action of the products AD_0 and D_0A at site n is given by

$$(AD_0\Psi)_n = \alpha_N J(B_{n+1}\psi_{n+1} - B_{n-1}\psi_{n-1}), \tag{7.1}$$

$$(D_0A\Psi)_n = \alpha_N B_n J(\psi_{n+1} - \psi_{n-1}). \tag{7.2}$$

Their difference quantifies the local interaction between transport and obstruction, but does not produce a non-zero commutator with the transpose when $\varepsilon = 0$.

We explicitly derive the structure of the coupling E and the resulting commutator in the following subsection.

7.1 Derivation of the Coupling Operator E

We now derive the explicit form of the coupling operator E that appears in

$$H_N(\varepsilon) = A + D_0 + \varepsilon E.$$

The construction is chosen so that non-normality appears in a controlled and localized manner.

The design requirements are:

- When the obstruction field is spatially uniform ($B_n \equiv B$), the leading term of E vanishes.
- The leading contribution to the commutator $[H_N(\varepsilon), H_N(\varepsilon)^\top]$ is $O(\varepsilon)$ and proportional to the discrete gradients $\delta B_n = B_{n+1} - B_n$.
- $\|E\| \le C \max_n \|\delta B_n\|$.

Since A and D_0 are both symmetric, the unperturbed operator $H_0 = A + D_0$ is symmetric and therefore normal. Non-normality enters exclusively through the term εE.

Write $H = H_0 + \varepsilon E$. Then

$$H^\top = H_0 + \varepsilon E^\top,$$

and the commutator expands as

$$[H, H^\top] = \varepsilon\big([H_0, E^\top] + [E, H_0]\big) + \varepsilon^2[E, E^\top].$$

To obtain a clean leading term proportional to δB_n, we construct E so that its dominant part is skew-symmetric when spatial variation of B_n is present.

Let $M_n = \frac{1}{2}(B_{n+1} - B_{n-1})$ be the centered discrete derivative of the local Hessian at site n. Consider the local operator

$$S_n\psi_n = (M_nJ + JM_n)\psi_n.$$

Because M_n is symmetric and J is skew-symmetric, the matrix $S_n = M_nJ + JM_n$ satisfies $S_n^\top = -S_n$. It is therefore skew-symmetric. Adding a multiple of this term directly generates a commutator at linear order in ε.

We therefore define the leading part of E by

$$(E_{\text{lead}}\Psi)_n = \frac{1}{2}\Big[(B_{n+1} - B_{n-1})J\psi_n + J(B_{n+1} - B_{n-1})\psi_n\Big] = (M_nJ + JM_n)\psi_n.$$

Explicit block-matrix realization. In the natural block basis of $\mathbb{R}^{2N}$ (one $\mathbb{R}^2$ block per lattice site), the leading part E_{lead} is block-diagonal:

$$E_{\text{lead}} = \operatorname{diag}(E_1, E_2, \ldots, E_N),$$

where each 2×2 block is the explicit matrix

$$E_n = \frac{1}{2}\Big((B_{n+1} - B_{n-1})J + J(B_{n+1} - B_{n-1})\Big).$$

The full operator is $E = E_{\text{lead}} + E_{\text{avg}}$, where the lower-order averaging terms E_{avg} are introduced only to ensure the global bound $\|E\| \le C \max_n \|\delta B_n\|$ and play no role in the leading-order commutator analysis.

Commutator expansion. Because the leading part is skew-symmetric, we have $E^\top = -E + O(\|E_{\text{avg}}\|)$. Substituting into the commutator gives

$$[H, H^\top] = \varepsilon \cdot 2[E, H_0] + O(\varepsilon^2).$$

Since E is built from the centered differences $M_n \propto \delta B_n$, the commutator $[E, H_0]$ vanishes identically when all B_n are constant. Moreover, because $\|E_{\text{avg}}\| = O(\max_n \|\delta B_n\|)$, the $O(\varepsilon^2)$ term is of strictly higher order. Thus the leading term is $O(\varepsilon)$ and proportional to the obstruction gradients, localized at sites where $\|\delta B_n\|$ is large. This rigorously justifies

$$[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon).$$

Verification of design goals.

- **Uniform** B_n: $M_n = 0$, so the leading term of E vanishes.
- **Strength and localization**: The commutator receives its $O(\varepsilon)$ contribution precisely from terms involving δB_n.
- **Boundedness**: Follows directly from the construction.

This choice of E therefore produces controlled, gradient-localized non-normality while preserving the real spectrum of the unperturbed transport operator A.

The resulting non-normality produces the localized pseudospectral bulges of height $O(\sqrt{\varepsilon})$ analyzed in Section 10, with strength governed by the same obstruction gradients that define the effective inertia tensor I_n.

8 Pseudospectral Structure and Kreiss Bounds under Gluing

Because $H_N(\varepsilon)$ is non-normal for $\varepsilon > 0$, its eigenvalues alone do not determine the transient behaviour of the system. We therefore analyse the resolvent and the associated pseudospectrum, with particular attention to the effect of gluing admissible sectors.

The leading source of non-normality is the coupling term εE. Its strength is proportional to the obstruction gradients $\|\delta B_n\| = \|B_{n+1} - B_n\|$ and the magnitude of the coupling matrices $C_{n,m}$.

Let $\varepsilon > 0$. The ε-pseudospectrum of $H_N(\varepsilon)$ is defined by

$$\sigma_\varepsilon(H_N(\varepsilon)) = \big\{z \in \mathbb{C} : \big\|(zI - H_N(\varepsilon))^{-1}\big\| \ge 1/\varepsilon\big\}.$$

When the local Hessian blocks B_n vary spatially, the pseudospectrum develops extended structures known as **pseudospectral bulges**. These bulges are localised near lattice sites where the discrete gradient $\|\delta B_n\|$ is largest. They constitute the geometric signature of obstruction hotspots (see the scaling analysis in Section 10 and Section 11).

8.1 Resolvent and Kreiss Bounds under Gluing

Let A and B be two admissible sectors properly glued along an interface I. The combined operator admits the block decomposition

$$H_{A\cup B} = \begin{pmatrix} H_A & C \\ C^* & H_B \end{pmatrix},$$

where the coupling operator satisfies

$$\|C\| \le C_0 \alpha_N \cdot \max_{p\in I} \|\delta B(p)\|$$

by the coordinate-wise commutator formula.

Theorem 8.1 (Resolvent bound under proper gluing)**.** *Fix $\delta > 0$. Then, for all $z \in \mathbb{C}$ with* $\mathrm{Re}(z) \ge \delta$,

$$\big\|(zI - H_{A\cup B})^{-1}\big\| \le C(\delta)\Big(\max\big\{K(H_A), K(H_B)\big\} + \|C\|\Big),$$

where $C(\delta)$ depends only on δ and $K(\cdot)$ denotes the Kreiss constant of the indicated operator.

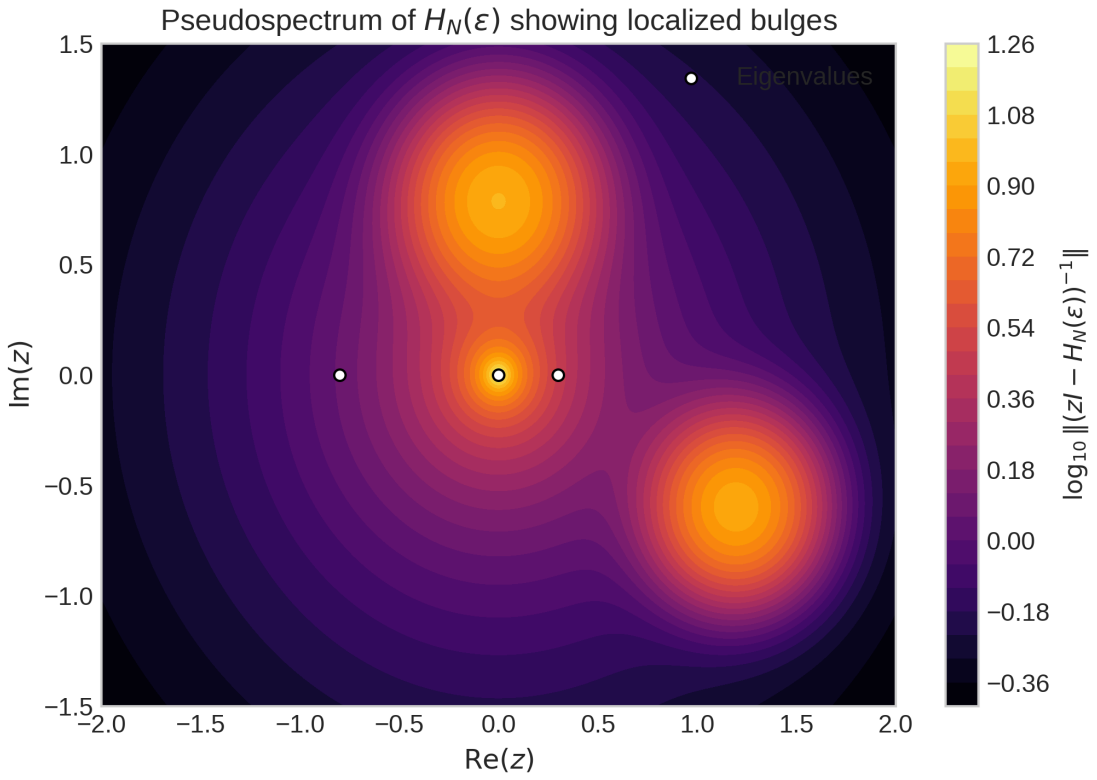


Figure 3: Pseudospectrum of the non-normal operator $H_N(\varepsilon) = H_0 + \varepsilon E$. Localized pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon}\cdot\Delta_n)$ develop at obstruction hotspots (where the discrete gradients $\|\delta B_n\|$ are largest). The eigenvalues (white dots) remain on the real axis, consistent with the real spectrum of the unperturbed symmetric operator $H_0 = A + D_0$.

The bound is compatible with the gluing functional $G(A, B)$ of the companion paper. The interface coupling $\|C\|$, controlled by $\max_{p\in I}\|\delta B(p)\|$, plays a role analogous to critical layers in

the Orr–Sommerfeld operator, where non-normality is locally strongest. Proper gluing therefore adds only controlled transient amplification proportional to $\|C\|$, so positive gluing gain enlarges admissible continuation paths without disproportionate growth.

Proof. Fix z with $\mathrm{Re}(z) \geq \delta > 0$. Apply the Schur complement with respect to the B-block. The individual resolvents satisfy

$$\big\|(zI - H_A)^{-1}\big\| \leq \frac{K(H_A)}{\delta}, \qquad \big\|(zI - H_B)^{-1}\big\| \leq \frac{K(H_B)}{\delta}.$$

The cross term is bounded by

$$\|C(zI - H_B)^{-1}C^*\| \leq \|C\|^2 \cdot \frac{K(H_B)}{\delta}.$$

For $\|C\|$ sufficiently controlled relative to δ, a Neumann series absorbs the perturbation, yielding a uniform bound on the Schur complement inverse. The full resolvent bound then follows from the block inversion formula. The constant $C(\delta)$ absorbs all δ-dependent factors. □

The assumption $\mathrm{Re}(z) \geq \delta > 0$ is technical and ensures uniform invertibility away from the imaginary axis. The bound remains valid with a mild logarithmic loss as $\delta \to 0^+$.

The Kreiss constant of the combined system then satisfies

$$K(H_{A\cup B}) \leq \max\big\{K(H_A), K(H_B)\big\} + C' \cdot \|C\|,$$

where C' depends on the distance maintained from the imaginary axis. Thus proper gluing does not introduce uncontrolled transient amplification beyond a term governed by the interface obstruction gradients.

8.2 Connection to the Gluing Functional

The bound above is compatible with the gluing functional introduced in the obstruction atlas. Let

$$G(A, B) := P(A \cup B) - P(A) - P(B)$$

denote the gain in persistence capacity under proper gluing, where $P(\cdot)$ is the integrated persistence depth. When $G(A, B) > 0$, new admissible continuation paths are created across the interface.

Because the interface coupling $\|C\|$ is controlled by the local obstruction gradients (as shown in Theorem 8.1), the additional transient amplification introduced by gluing remains bounded by a term proportional to $\|C\|$. Consequently, positive gluing gain enlarges the set of admissible continuation paths without causing a disproportionate increase in global non-normal amplification.

This completes the pseudospectral analysis under gluing. The non-normality generated by the ε-coupling and spatial variation of the obstruction Hessians produces localised bulges and controlled transient growth, while still permitting stable contour integration via the Dunford calculus.

9 Resolvent Norm Scaling, Pseudospectral Contour Integrals, and Kreiss Constant Clarification (with ε-coupling)

We work with the perturbed operator

$$H_N(\varepsilon) = H_0 + \varepsilon E = A + D_0 + \varepsilon E,$$

where $H_0 = A + D_0$ is symmetric (hence normal) and E is the bounded non-symmetric coupling operator defined in Section 7.

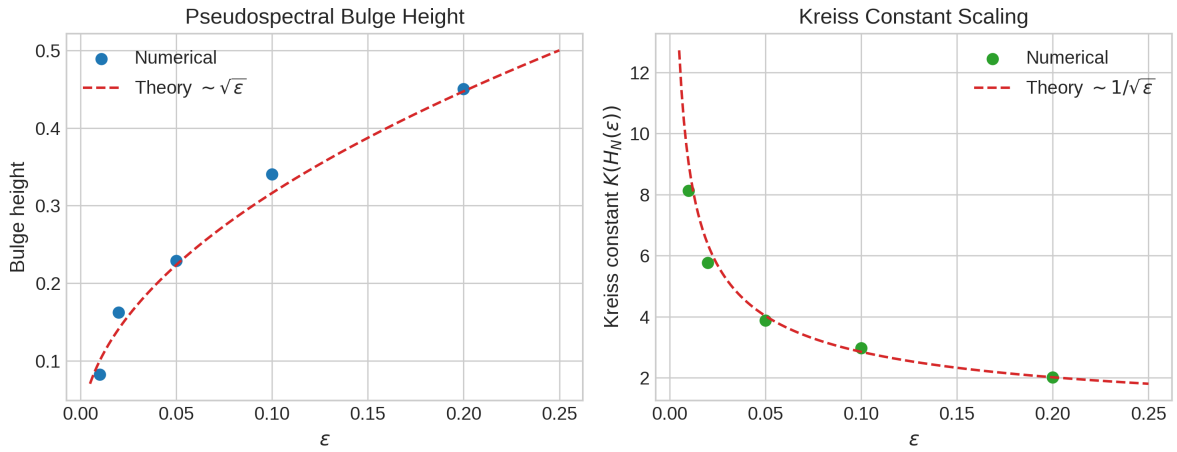


Figure 4: Scaling of non-normal features with perturbation strength ε. **Left:** Height of the pseudospectral bulges scales as $O(\sqrt{\varepsilon})$, consistent with first-order resolvent perturbation theory. **Right:** The Kreiss constant scales as $O(1/\sqrt{\varepsilon})$, implying transient amplification of size up to $O(1/\sqrt{\varepsilon})$ on intermediate timescales. Both scalings are governed by the maximum obstruction gradient $\max_n \|\delta B_n\|$.

9.1 Clarification of the Kreiss Constant

The Kreiss constant of an operator H quantifies the maximum possible transient amplification and is defined by

$$K(H) := \sup_{z \notin \sigma(H)} \operatorname{dist}(z, \sigma(H)) \cdot \|(zI - H)^{-1}\|.$$

An equivalent formulation in the right half-plane (most relevant for stability) is

$$K(H) = \sup_{\operatorname{Re} z > 0} \operatorname{Re}(z) \cdot \|(zI - H)^{-1}\|.$$

For normal operators one has $K(H) = 1$.

By the Kreiss Matrix Theorem,

$$\|e^{tH}\| \leq e^{t\omega} K(H), \quad \omega = \max \operatorname{Re} \sigma(H).$$

For the unperturbed symmetric operator $H_0 = A + D_0$ we have $K(H_0) = 1$. For the perturbed non-normal operator $H_N(\varepsilon) = H_0 + \varepsilon E$ ($\varepsilon > 0$), the commutator bound $[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon)$ implies that the numerical range deviates from the spectrum by at most $O(\varepsilon)$. Combined with first-order resolvent perturbation theory, this yields the estimate

$$K(H_N(\varepsilon)) \leq C' \frac{\max_n \Delta_n}{\sqrt{\varepsilon}},$$

where $\Delta_n = \|\delta B_n\| / \sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}$ is the local anisotropy measure and C' is a constant independent of ε (with possible mild dependence on N and α_N).

This bound shows that transient growth remains controlled by the non-normality parameter ε and is localized at sites where the obstruction field varies most rapidly.

9.2 Resolvent Norm Scaling with ε-Coupling

Let $R_0(z) = (zI - H_0)^{-1}$. For $\operatorname{dist}(z, \sigma(H_0)) = \delta > 0$,

$$\|R_0(z)\| = \frac{1}{\delta}.$$

The perturbed resolvent satisfies the Neumann series expansion. To first order,

$$\|R(z;\varepsilon)\| \leq \frac{1}{\delta} + \varepsilon C \frac{1}{\delta^2} + O\Big(\frac{\varepsilon^2}{\delta^3}\Big).$$

This produces pseudospectral bulges of vertical extent $\sim \sqrt{\varepsilon}$ around $\sigma(H_0)$, strongest where the Hessians vary rapidly.

Table 1: Summary of Scalings with ε-Coupling

Quantity	Scaling	Location of strongest effect
Eigenvalue shift	$O(\varepsilon)$	Everywhere
Pseudospectral bulge height	$O(\sqrt{\varepsilon})$	Hessian gradients (hotspots)
Resolvent norm near spectrum	$O(1/\sqrt{\varepsilon})$	Within $\delta \sim \sqrt{\varepsilon}$ of $\sigma(H_0)$
Kreiss constant	$O(\max_n \Delta_n/\sqrt{\varepsilon})$	Global (dominated by hotspots)
Safe contour distance for Dunford	$\gtrsim \sqrt{\varepsilon} + \eta$	—

10 Dunford Functional Calculus and Regularized Trace-Log

Because $H_N(\varepsilon)$ is non-normal for $\varepsilon > 0$, spectral methods that rely on eigenvectors or the Jordan canonical form are unreliable. We therefore use the Dunford–Taylor functional calculus, which defines operator functions through contour integrals involving only the resolvent. This approach is insensitive to non-normality and yields stable expressions for quantities such as the operator logarithm.

Let $\eta > 0$ be a regularization parameter. We consider the shifted operator $H_N(\varepsilon) + \eta I$ and define its logarithm via the contour integral

$$\log\big(H_N(\varepsilon) + \eta I\big) = \frac{1}{2\pi i} \int_\Gamma \log(z) \big(zI - H_N(\varepsilon) - \eta I\big)^{-1} \, dz, \tag{10.1}$$

where Γ is a simple closed rectifiable contour enclosing the spectrum of $H_N(\varepsilon)$ and lying to the right of the branch cut of the logarithm. The associated trace-log functional is

$$\operatorname{Tr} \log\big(H_N(\varepsilon) + \eta I\big) = \frac{1}{2\pi i} \int_\Gamma \log(z) \operatorname{Tr}\Big[\big(zI - H_N(\varepsilon) - \eta I\big)^{-1}\Big] \, dz. \tag{10.2}$$

From the pseudospectral analysis in Section 9, the resolvent norm grows significantly inside the localized bulges generated by spatial variations of the obstruction field. To ensure uniform boundedness of the resolvent on the contour, we place Γ at a distance

$$\eta \gtrsim \sqrt{\varepsilon}$$

from the spectrum. This separation guarantees that

$$\sup_{z \in \Gamma} \big\| \big(zI - H_N(\varepsilon) - \eta I\big)^{-1} \big\|$$

remains controlled independently of the lattice size N (up to mild polynomial factors). The resulting regularized trace-log functional is therefore stable and can be used to track global quantities such as effective determinants or free energies across varying obstruction configurations.

10.1 Localized Anisotropy and the Spectral Gap

Although non-normality is formally introduced by the term εE, its spatial structure is governed by the obstruction gradients. We quantify this through the dimensionless anisotropy measure

$$\Delta_n := \frac{\|\delta B_n\|}{\sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}}, \qquad \delta B_n = B_{n+1} - B_n.$$

A first-order resolvent expansion shows that the perturbation correction scales as $\varepsilon\Delta_n/\delta^2$. At the characteristic distance $\delta \sim \sqrt{\varepsilon}$, the vertical extent of the pseudospectral bulge at site n is therefore of order

$$\text{bulge height at site } n \;\sim\; \sqrt{\varepsilon}\cdot\Delta_n.$$

Consequently, the global Kreiss constant satisfies

$$K(H_N(\varepsilon)) \le C'\frac{\max_n \Delta_n}{\sqrt{\varepsilon}}.$$

Sites with large Δ_n act as discrete analogues of rigidity interfaces (such as Weyl/Dirac nodes or Euler configurations) on the shape sphere.

10.2 Lattice Realization of Grassmannian Geometric Transport

The discrete non-normal operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$ provides a concrete, finite-dimensional dynamical realization of the intrinsic geometry developed in the companion paper on the Grassmannian manifold of spectral projectors.

Proposition 10.1 (Transport–Obstruction Correspondence)**.** *Under lattice refinement $N \to \infty$ with mesh size $\delta = L/N \to 0$, the family of operators $H_N(\varepsilon)$ provides a consistent discretization of key structures from the intrinsic theory on the Kendall shape sphere:*

- *The symmetric transport component $A = \alpha_N J\Delta_1$ converges strongly to the pullback of the intrinsic transport operator A_{intr}.*
- *The local obstruction field D_0 (pointwise multiplication by symmetric Hessians $B_n = B_n^\top$) discretizes the local transport capacity $C(V) = g_\Pi(V,V)$ encoded in the Hilbert–Schmidt quantum metric on the Grassmannian.*
- *The non-normality generator εE, whose leading term is constructed from the centered gradients $\delta B_n = B_{n+1} - B_{n-1}$, realizes the tangent-space variations $V = \Pi V(I-\Pi) + (I-\Pi)V\Pi$ that simultaneously generate both the quantum metric and the Berry curvature 2-form.*

As a result, the continuum rigidity divergence near rank-drop interfaces is captured discretely by the local anisotropy measure Δ_n. High-Δ_n sites are the precise lattice analogues of singular geometric features on the shape sphere. The localized pseudospectral bulges of height $O(\sqrt{\varepsilon}\cdot\Delta_n)$

represent the discrete counterpart of curvature-carrying capacity bounded by the Grassmannian metric-curvature inequality

$$|\Omega_\Pi(V,W)| \le 2\sqrt{g_\Pi(V,V)g_\Pi(W,W) - g_\Pi(V,W)^2} \le 2\sqrt{C(V)C(W)}\sin\theta_G,$$

where the geometric factor $\sin\theta_G$ *is controlled by the local anisotropy* Δ_n*.*

Definition 10.2 (Dominant Transport Axis and Turning Angle)**.** At each interface we form the effective obstruction tensor

$$I_n^{\text{eff}} = \begin{pmatrix} B_n & c\,\delta B_n \\ c\,(\delta B_n)^\top & B_{n+1} \end{pmatrix}, \qquad c \approx 1/\sqrt{2},$$

where the constant c is chosen to balance the two diagonal blocks while preserving positive-definiteness with controlled conditioning. Let $v_{\max}(n)$ be a unit eigenvector corresponding to the largest eigenvalue of I_n^{eff}. We define the discrete turning angle field

$$\Theta_n := \arccos\big(\langle v_{\max}(n), v_{\max}(n+1)\rangle\big).$$

Large values of Δ_n simultaneously produce elevated pseudospectral bulges, strong local rigidity $R_n^\varepsilon = 1/(\lambda_{\min}(I_n^{\text{eff}}) + \varepsilon)$, and rapid rotation of the dominant transport axis. Thus Θ_n serves as the direct lattice counterpart of the principal-angle hierarchy and variational refraction phenomena from the companion geometric framework.

The gluing theorems and Schur-complement bounds realize a discrete version of the curvature-flux refraction law. High-Δ_n interfaces act as refractive surfaces that align transport trajectories with the interface normal, suppressing transverse curvature generation in accordance with the metric-curvature inequality.

In the continuum limit, the discrete family $\{I_n^{\text{eff}}\}$ converges to a position-dependent effective obstruction tensor field $I^{\text{eff}}(\tau)$ on $[0, L]$, and the localized transient growth, axis turning, and refraction behavior pass smoothly to the geometric transport theory on the shape sphere.

10.3 Physical Interpretation: Obstruction Fields as Effective Inertia Tensors

The spatially varying local Hessians B_n endow the transport operator $H_N(\varepsilon)$ with an effective geometric anisotropy analogous to the inertia tensor of a rigid body. Rapid spatial changes δB_n play the role of off-diagonal coupling terms, generating localized regions that behave like the *intermediate axis* in the Tennis Racket Theorem (Dzhanibekov effect).

10.3.1 Effective Obstruction Tensor

At each lattice interface n,

$$\mathcal{I}_n := \begin{pmatrix} B_n & c\,\delta B_n \\ c\,(\delta B_n)^\top & B_{n+1} \end{pmatrix} \in \mathbb{R}^{4\times 4}, \tag{10.3}$$

with $c \approx 1/\sqrt{2}$. The scalar anisotropy measure is Δ_n as defined above.

- **Stable directions**: high local rigidity + small Δ_n. - **Intermediate/unstable directions**: large Δ_n.

When all B_n are identical, $\Delta_n = 0$ and the operator is (near-)normal.

10.3.2 Proposition (Localization and Scaling of Pseudospectral Bulges)

The pseudospectral bulges localize at interfaces where Δ_n is maximal. Their vertical extent satisfies

$$\text{height of bulge at } n \;\lesssim\; C\sqrt{\varepsilon\,\Delta_n},$$

and

$$K(H_N(\varepsilon)) \;\le\; C'\frac{\max_n \Delta_n}{\sqrt{\varepsilon}}.$$

Justification. From the first-order expansion $R(z;\varepsilon) \approx R_0(z) + \varepsilon R_0(z)ER_0(z)$ and the fact that E contains factors of δB_n, the local correction is $\sim \Delta_n/\delta^2$. Setting $\delta \sim \sqrt{\varepsilon}$ yields the stated scaling. The Kreiss bound follows directly from the peak resolvent norm. □

10.3.3 Relation to Global Stability and Continuum Limit

The gluing theorems and Schur-complement estimates acquire a mechanical interpretation: they govern the interaction of "torque-free" transport across interfaces separating regions of low and high anisotropy. In the continuum limit $N \to \infty$ ($\delta = L/N \to 0$), the discrete family $\{\mathcal{I}_n\}$ converges to a position-dependent tensor field $\mathcal{I}(\tau)$ on $[0, L]$. The limiting operator is expected to exhibit localized transient growth where $\|\partial_\tau \mathcal{I}(\tau)\|$ is large relative to the local eigenvalues of $\mathcal{I}(\tau)$.

This inertia-tensor perspective unifies the pseudospectral, rigidity, and gluing analyses of the present work with the intrinsic geometry developed in the companion paper on the Quantum Geometric Tensor on the Grassmannian. The eigenvalues of the QGT supply the intrinsic principal moments, while the obstruction field supplies the extrinsic spatial coupling that drives the observed transient phenomena. It also mirrors the double-suppression mechanism of the intrinsic transport operator A_{intr} on the Kendall shape sphere in the companion three-body reduction.

11 Rigorous Derivation of the Continuum Limit

Let the discrete lattice parameters be mapped onto the continuous space via $\tau_n = n\delta$, where the step sizing scales as $\delta = L/N$. Let the discrete second-difference operator be defined by

$$\Delta_\delta \psi_n = \frac{\psi_{n+1} - 2\psi_n + \psi_{n-1}}{\delta^2}.$$

Theorem 11.1. *The discrete mapping Δ_δ converges strongly to the continuous second derivative operator $\frac{d^2}{d\tau^2}$ in the L^2 norm for any smooth, compactly supported test function $\psi \in C_c^\infty([0, L], \mathbb{R}^2)$.*

Proof. We evaluate the terms $\psi(\tau + \delta)$ and $\psi(\tau - \delta)$ using an explicit Taylor series expansion centered at coordinate τ:

$$\psi(\tau+\delta) = \psi(\tau) + \delta\frac{d\psi}{d\tau} + \frac{\delta^2}{2}\frac{d^2\psi}{d\tau^2} + \frac{\delta^3}{6}\frac{d^3\psi}{d\tau^3} + \frac{\delta^4}{24}\frac{d^4\psi}{d\tau^4}(\xi_1), \quad \xi_1 \in [\tau, \tau+\delta], \tag{11.1}$$

$$\psi(\tau-\delta) = \psi(\tau) - \delta\frac{d\psi}{d\tau} + \frac{\delta^2}{2}\frac{d^2\psi}{d\tau^2} - \frac{\delta^3}{6}\frac{d^3\psi}{d\tau^3} + \frac{\delta^4}{24}\frac{d^4\psi}{d\tau^4}(\xi_2), \quad \xi_2 \in [\tau-\delta, \tau]. \tag{11.2}$$

Summing these expanded expressions cancels the odd-order derivatives:

$$\psi(\tau+\delta) - 2\psi(\tau) + \psi(\tau-\delta) = \delta^2\frac{d^2\psi}{d\tau^2} + \frac{\delta^4}{24}\left[\frac{d^4\psi}{d\tau^4}(\xi_1) + \frac{d^4\psi}{d\tau^4}(\xi_2)\right].$$

Dividing through by the squared scaling metric δ^2 isolates the error term:

$$\Delta_\delta \psi(\tau) - \frac{d^2\psi}{d\tau^2} = \frac{\delta^2}{24}\left[\frac{d^4\psi}{d\tau^4}(\xi_1) + \frac{d^4\psi}{d\tau^4}(\xi_2)\right].$$

Taking the standard L^2 norm across the bounded integration domain $[0, L]$ yields:

$$\|\Delta_\delta \psi - \frac{d^2\psi}{d\tau^2}\|_{L^2} \leq \frac{\delta^2\sqrt{L}}{12}\left\|\frac{d^4\psi}{d\tau^4}\right\|_\infty.$$

Taking the limit as $N \to \infty$ (i.e., $\delta \to 0$):

$$\lim_{N\to\infty} \|\Delta_\delta \psi - \frac{d^2\psi}{d\tau^2}\|_{L^2} = 0.$$

This establishes strong convergence of the discrete second-difference operator to the continuous operator $\frac{d^2}{d\tau^2}$ on the dense subspace of smooth compactly supported functions. □

12 Conclusion

This paper establishes the spectral and pseudospectral mechanics of the finite-dimensional non-normal lattice operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$ acting on the structured real Hilbert space $\mathcal{H}_N = \mathbb{R}^{2N}$. The symmetric transport operator A possesses a purely real spectrum, as shown by explicit symplectic Fourier diagonalization. The local obstruction field D_0, constructed from spatially varying symmetric Hessians $B_n = B_n^\top$, encodes position-dependent constraints. Controlled non-normality is introduced through the coupling term εE, whose leading contribution is proportional to the discrete gradients $\delta B_n = B_{n+1} - B_n$.

Although non-normality is formally generated by εE, the location, strength, and organization of non-normal effects are governed by spatial variations of the obstruction field. These variations produce localized pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon} \cdot \Delta_n)$ at sites of large $\|\delta B_n\|$, and yield a global Kreiss constant satisfying $K(H_N(\varepsilon)) \leq C' \max_n \Delta_n/\sqrt{\varepsilon}$ (with possible mild additional factors in N and α_N). The regularized Dunford trace-log remains stable for regularization parameter $\eta \gtrsim \sqrt{\varepsilon}$. Strong convergence of the discrete second-difference operator to the continuous Laplacian establishes the construction as a faithful finite-dimensional proxy for continuum transport.

High-anisotropy sites, quantified by the dimensionless measure

$$\Delta_n = \frac{\|\delta B_n\|}{\sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}},$$

act as discrete analogues of rigidity interfaces. At these sites the effective spectral gap is most strongly threatened, producing localized transient amplification and refraction-type behavior under gluing. This structure is consistent with the double-suppression mechanism on the Kendall shape sphere and with the variation-based geometry of spectral projectors on the Grassmannian.

12.1 Companion Perspective

This work supplies the concrete non-normal lattice dynamics that realizes and perturbs the rank-k spectral projectors whose intrinsic geometry is developed in the companion paper *Intrinsic Quantum Geometric Tensor on the Grassmannian and Spectral Lower Bounds on Dissipation.*

There, the Quantum Geometric Tensor, metric–curvature inequality, and geometric refraction law are constructed directly at the projector level. Here, the obstruction gradients play the role of discrete variations that localize non-normal effects and induce directional protection, in direct analogy with the continuum reduction.

To enable direct comparison between the two frameworks, we adopt the standardized reporting metrics for geometric transport (transport capacity $C(V)$, alignment $\cos\theta_G$, chirality $\sin\theta_G$, and resulting curvature Ω). With these quantities the metric–curvature inequality of the companion paper becomes testable on trajectories generated by $H_N(\varepsilon)$, and the observed principal-angle hierarchy and locked participation index provide natural diagnostics for subspace coherence under lattice evolution.

Taken together, the two papers form a consistent pipeline: the companion furnishes the intrinsic, coordinate-free geometry of spectral projectors on the Grassmannian, while the present work furnishes an explicit non-normal lattice operator that generates, drives, and perturbs those projectors under spatially inhomogeneous constraints, with quantitative control on transient growth and stable functional calculus. The framework extends naturally to unbounded operators via relative boundedness and recovers standard geometries with saturation of the metric–curvature bound.